

# Hyper SuperServer SYS-620H-TN12R

2U Hyper with 12 NVMe/SAS/SATA bays and up to 8 PCIe slots with GPU support



## Key Applications

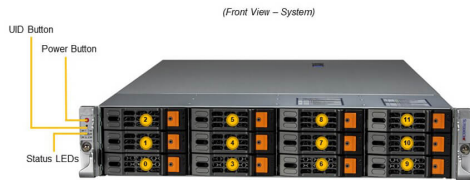
Virtualization, Software-defined Storage, AI Inference and Machine Learning,  
Cloud Computing, Enterprise Server,

## Key Features

- Dual Socket P+ (LGA-4189) 3rd Gen Intel® Xeon® Scalable Processors;
- 32 DIMM Slots; Up to 8TB DRAM; Up to 8TB Intel® Optane™ Persistent Memory (up to 12TB with DRAM); 3200/2933/2666 ECC DDR4 LRDIMM; RDIMM; Intel® Optane™ Persistent Memory 200 series;
- Optional PCIe slot configurations up to 8 PCIe 4.0 x8 or 4 PCIe 4.0 x16 slots with support for double-width GPU/Accelerator cards;
- Flexible networking options with up to 2 AIOM networking slots (OCP NIC 3.0 compatible);
- 4 heavy duty hot-swap fans with optimal fan speed control;



|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form Factor              | 2U Rackmount<br>Enclosure: 437 x 88.9 x 803mm (17.2" x 3.5" x 31.6")<br>Package: 605 x 263 x 1107mm (23.8" x 10.4" x 43.6")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Processor                | Dual Socket P+ (LGA-4189)<br>3rd Gen Intel® Xeon® Scalable processors<br>Up to 40C/80T; Up to 60MB Cache per CPU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| GPU                      | Up to 4 double-width GPUs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| System Memory            | Slot Count: 32 DIMM slots<br>Max Memory (2DPC): Up to 8TB 3200MT/s ECC DDR4<br>Supports Intel® Optane™ persistent memory 200 series                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Drive Bays Configuration | Default: Total 12 bays <ul style="list-style-type: none"> <li>• 12 front hot-swap 3.5" NVMe*/SAS*/SATA* drive bays</li> </ul> (*NVMe/SAS/SATA support may require additional storage controller and/or cables)<br>M.2: 2 M.2 NVMe/SATA slots (M-key 2280/22110)                                                                                                                                                                                                                                                                                                                                                             |
| Expansion Slots          | Default* <ul style="list-style-type: none"> <li>• 8 PCIe 4.0 x8 (in x16) FH/10.5"L slots</li> <li>• 2 PCIe 4.0 x16 AIOM slots (OCP 3.0 compatible)</li> </ul> Option A* <ul style="list-style-type: none"> <li>• 2 PCIe 4.0 x8 (in x16) FH/10.5"L slots</li> <li>• 3 PCIe 4.0 x16 FH/10.5"L slots</li> <li>• 2 PCIe 4.0 x16 AIOM slots (OCP 3.0 compatible)</li> </ul> (*Requires additional parts, please see the optional parts list for details. For more details on PCIe slot configuration options, please refer to the system callout images above.)                                                                  |
| On-Board Devices         | SATA: SATA (6Gbps) ; RAID 0/1/5/10 support<br>NVMe: NVMe; RAID 0/1/5/10 support(Intel® VROC RAID key required)<br>Network Connectivity: 2 RJ45 1GbE with Intel® I350-AM2 (optional)<br>4 RJ45 1GbE with Intel® I350-AM4 (optional)<br>4 SFP 1GbE with Intel® I350-AM4 (optional)<br>2 RJ45 10GBASE-T with Intel® X550-AT2 (optional)<br>2 SFP+ 10GbE with Intel® X710-BM2 (optional)<br>4 SFP+ 10GbE with Intel® XL710-BM1 (optional)<br>4 SFP+ 10GbE with Intel® X710-TM4 (optional)<br>2 SFP28 25GbE with Broadcom® BCM57414 (optional)<br>4 SFP28 25GbE (optional)<br>2 QSFP28 100GbE with Broadcom® BCM57508 (optional) |



| Slot    | Description                                                        |
|---------|--------------------------------------------------------------------|
| 1 - 12  | 3.5" or 2.5" Hot-Swap NVMe/SAS3/SATA3 Drive Bays (NVMe from CPU1)* |
| 13 - 24 | 3.5" or 2.5" Hot-Swap NVMe/SAS3/SATA3 Drive Bays (NVMe from CPU2)* |

\*NVMe, SAS3, or SATA3 support requires additional parts in optional parts list



| Slot Description | Option 1*             | Option 2*     | Option 3*                                |
|------------------|-----------------------|---------------|------------------------------------------|
| 1                | PCI-E 4.0 x8 (in x16) | PCI-E 4.0 x16 | 2x Hot-Swap 2.5" NVMe Drive Bays (CPU 1) |
| 2                | PCI-E 4.0 x8 (in x16) | —             | —                                        |
| Slot Description | Option 1*             | Option 2*     | Option 3*                                |
| 3                | PCI-E 4.0 x8 (in x16) | PCI-E 4.0 x16 | 2x Hot-Swap 2.5" NVMe Drive Bays (CPU 2) |
| 4                | PCI-E 4.0 x8 (in x16) | —             | —                                        |

\*Requires additional parts in optional parts list

| Slot Description | Option 1                          | Option 2*             |
|------------------|-----------------------------------|-----------------------|
| 5                | PCI-E 4.0 x8 (in x16)             | PCI-E 4.0 x16         |
| 6                | PCI-E 4.0 x8 (in x16)             | —                     |
| 7                | PCI-E 4.0 x8 (in x16)             | PCI-E 4.0 x8 (in x16) |
| 8                | PCI-E 4.0 x8 (in x16)             | PCI-E 4.0 x16         |
| 9                | PCI-E 4.0 x8 (in x16)             | —                     |
| Slot Description | Option 1                          | Option 2*             |
| A1               | AIO / OCP NIC 3.0 Slot            | —                     |
| A2               | (Optional) AIO / OCP NIC 3.0 Slot | —                     |

CPU1 CPU2

|                       |                                                                                                                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Cooling        | Fans: Up to 4x 8cm heavy duty fans with optimal fan speed control<br>Air Shroud: 2 Air Shrouds                                                                                                                                                                                   |
| Power Supply          | 2x 1200W Redundant Titanium Level (96%) Hot-plug power supplies                                                                                                                                                                                                                  |
| System BIOS           | BIOS Type: AMI 256MB SPI Flash                                                                                                                                                                                                                                                   |
| Management            | SuperCloud Composer®; Supermicro Server Manager (SSM); Supermicro Update Manager (SUM); Supermicro SuperDoctor® 5 (SD5); Super Diagnostics Offline (SDO); Supermicro Thin-Agent Service (TAS); SuperServer Automation Assistant (SAA) New!                                       |
| PC Health Monitoring  | CPU: Monitors for CPU Cores, Chipset Voltages, Memory<br>FAN: Fans with tachometer monitoring<br>Pulse Width Modulated (PWM) fan connectors<br>Status monitor for speed control<br>Temperature: Monitoring for CPU and chassis environment<br>Thermal Control for fan connectors |
| Dimensions and Weight | Weight: Gross Weight: 63 lbs (28.6 kg)<br>Net Weight: 37 lbs (16.8 kg)<br>Available Color: N/A                                                                                                                                                                                   |
| Operating Environment | Name: A2<br>Operating Temperature: 10°C to 35°C (50°F to 95°F)<br>Non-operating Temperature: -30°C to 60°C (-40°F to 140°F)<br>Operating Relative Humidity: 8% to 80% (max 21° DP; non-condensing)<br>Non-operating Relative Humidity: 8% to 90% (max 38° DP; non-condensing)    |
| Motherboard           | <a href="#">Super X12DHM-6</a>                                                                                                                                                                                                                                                   |
| Chassis               | CSE-HS829-R1K24P                                                                                                                                                                                                                                                                 |